SUMMARY

This data science case study was done for X Education, which sells online courses, and the objective was to find ways to get more leads to join their courses i.e., to identify *hot leads* who would most likely purchase a course, and thereby improve the lead conversion rate of the company. The steps taken are as follows -

1. Data Cleaning

The original dataset given to us was only partially clean, except for some missing values and inconsistencies in data formats. Initially, identification of variables with highest null value percentage was done, and if it was greater than 30%, they were dropped. There was a 'Select' level in some variables, which had to be replaced with a null value since it did not give us much information. Few other variables that had null values were imputed with either the median value (for numerical variables) or the most frequently occurring value (for categorical variables). There were no duplicates or many outliers. Hence this job was taken care of. There was one variable with inconsistent formats in its values, which was corrected.
Some other unimportant variables were dropped from the dataset altogether.

2. EDA – Univariate and Bivariate Analyses

A deep and extensive EDA was done to check the condition of our data. Univariate analyses of several variables were done. The objective was to understand the distribution of each variable in the dataset, and identify outliers if any. No outliers were found in the data. Followed by this were several bivariate analyses done to study the impact of each variable on the target variable – 'Converted'. Visualizations were plotted for each of these analyses.

3. Dummy Variables & Data Preparation

Dummy variables were created to convert the categorical variables into numerical ones for model building. Further preparation of data was done, and all unnecessary columns were deleted.

4. Train-Test Split

The split was done at 70% and 30% for train and test data respectively.

5. Feature Scaling & Model Building

First, RFE was done to attain the top 15 relevant variables. Later, the rest of the variables were removed manually depending on the VIF values and p-value (variables with VIF < 5 and p-value < 0.05 were kept).

6. Model Evaluation

A confusion matrix was made. Later, the optimum cut off value (using ROC curve) was used to find the accuracy, sensitivity and specificity which came to be around 80% each.

7. Prediction

Prediction was done on the test data frame and with an optimum cutoff as 0.358, and with accuracy, sensitivity and specificity of approximately 80% for both training and test datasets.

8. Conclusion

The variables that matter the most were found to be -

1. *Total time spent on the website*
2. *When their current occupation is a working professional*
3. *When the lead source was*
   - *a. Google*
   - *b. Direct Traffic*
   - *c. Organic search*
   - *d. Welingak website*
4. *When the last activity was*
   - *a. SMS*
   - *b. Olark chat conversation*
5. *When the lead origin is Lead add format*

Keeping these in mind, X Education can flourish if they focus on leads satisfying the above criteria. They will be able to effectively increase their lead conversion rate by a significant number.